



IILM University

School of Computer Science and Engineering SUMMER INTERNSHIP PRESENTATION

AI Automation and Full-Stack Software Development

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Enrollment No.: CS-2341726

B.Tech CSE | 7th Semester | 4CSE1

Organization: ScaleOn

Summer Internship Presentation | IILM University | Nitish Kumar Singh



Internship Overview

AI Automation

Summer Internship focused on practical AI automation and full-stack software development.

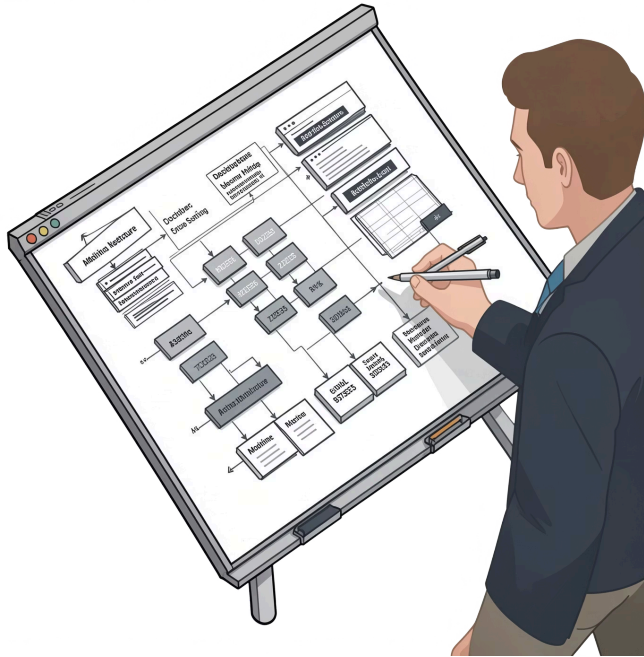
Three Major Projects

LinkedIn Integration Module, AI Email Automation System, and ScaleOn Study Portal.

Hands-On Exposure

APIs, OAuth 2.0, AI services, dashboards, databases, authentication and GitHub collaboration.

Objectives



1 Real-World Solutions

Build practical software solutions for real-world automation and platform-management problems.

2 Secure Integrations

Integrate external services securely through APIs and OAuth.

3 Generative AI

Use generative AI to automate email classification and response drafting.

4 Full-Stack Platform

Contribute to a full-stack internship management and learning platform.

Project 1 — LinkedIn Integration Module

FastAPI Backend

FastAPI backend for LinkedIn account integration and social-media publishing.

OAuth 2.0 Flow

OAuth 2.0 connection flow without handling user passwords.

Core Features

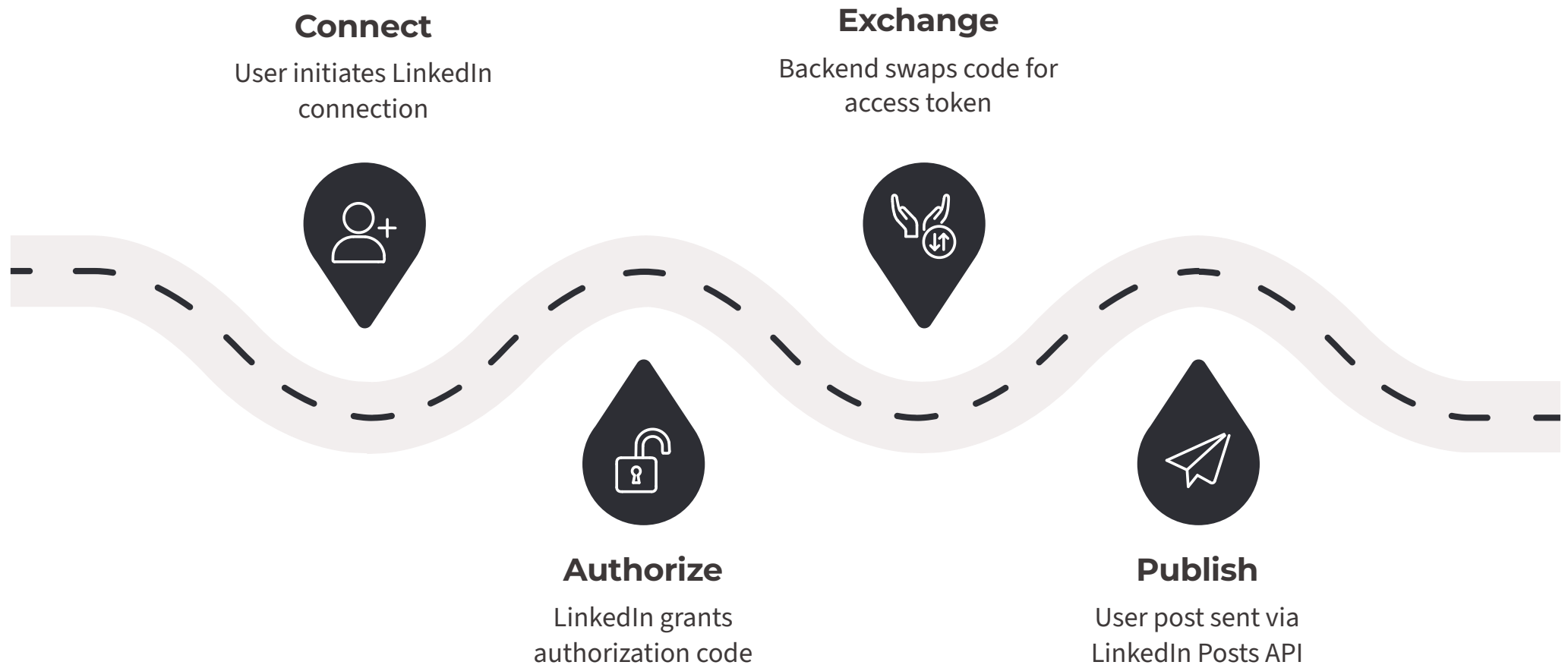
Profile retrieval, text posting, image posting and disconnect functionality.

Tech Stack

FastAPI, httpx, SQLAlchemy/SQLite, python-dotenv and Docker.



LinkedIn — Architecture & Workflow



The backend exchanges the authorization code for an access token, stores the connected profile and token, then uses the saved token to publish posts directly to LinkedIn.

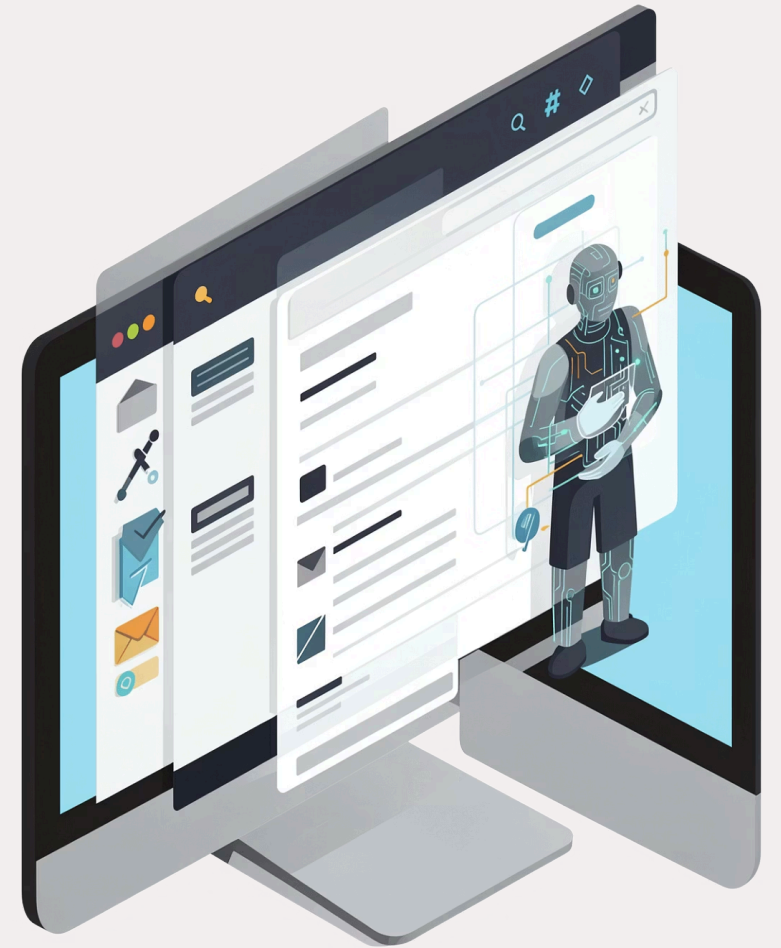
Project 2 — AI Email Automation System

AI-powered Gmail workflow developed for the ScaleOn AI Agent Developer track.

- Fetches all or unread emails and prevents duplicate processing using `message_id`.
- Gemini classifies category, priority, intent and sentiment.
- Generates context-aware professional replies.

Powered By

- Google Gemini API
- IMAP / SMTP
- Streamlit Dashboard
- CSV Logging



AI Email — Workflow



01

Fetch Emails

Gmail Inbox → IMAP email reader → duplicate check via `message_id`.

02

Classify

New email → Gemini classification → category / priority / intent / sentiment.

03

Generate Reply

AI generates suggested reply → processing result logged to CSV.

04

Human Review

Streamlit dashboard → human review / edit → Save as Draft or Send.

Project 3 — ScaleOn Study Portal

Full-stack internship management and learning platform.

Frontend

Next.js 16, React 19, TypeScript and Tailwind CSS 4.

Backend

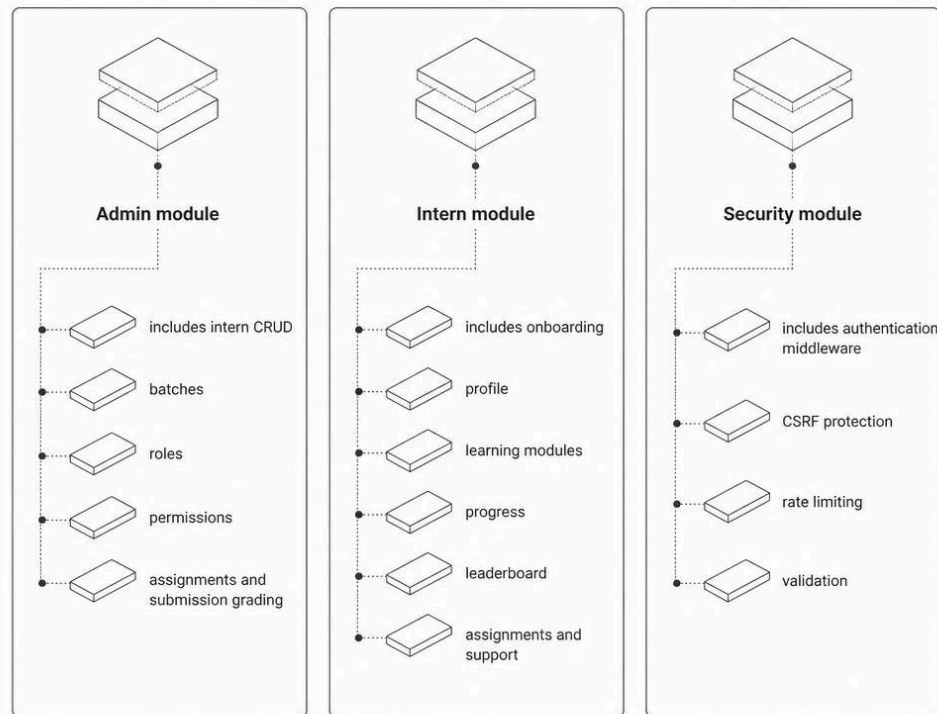
Node.js 22, Express.js and TypeScript.

Database

PostgreSQL with Prisma ORM.



Study Portal — Key Modules



Admin

Intern CRUD, batches, roles, permissions, assignments and submission grading.

Intern

Onboarding, profile, learning modules, progress, leaderboard, assignments and support.

Security

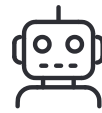
Authentication middleware, CSRF protection, rate limiting and validation.

Technologies & Tools



Backend & APIs

Python, FastAPI, Uvicorn, httpx, SQLAlchemy, SQLite, Docker.



AI & Data

Streamlit, Plotly, Google Gemini API, IMAP/SMTP, CSV logging.



Frontend

Next.js, React, TypeScript, Tailwind CSS, Node.js, Express.js.



Database & UI

PostgreSQL, Prisma ORM, JWT, Framer Motion and Lucide React.

Learning Outcomes & Challenges

Learning

OAuth 2.0, generative AI integration, REST APIs, authentication, databases and layered architecture.

Challenges Solved

API constraints, credential security, duplicate processing, human control in AI automation, complex workflows and team collaboration.

